Joann Ching

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Education

PhD in Computer Science, Johannes Kepler University, Linz, Austria *2024 - present*
- Computational Modeling and Analysis for Expressive Musical Performance

M.S. in Music Technology, Georgia Institute of Technology, USA *2021 - 2023*
- Master Project - Acoustic Analysis and Detection for Piano Sustain Pedal
- Relevant courses: DSP, Statistics for Machine Learning, Music Perception and Cognition

Professional Studies Program, The National Chiao Tung University, Taipei, Taiwan *2019 Fall*
- Relevant courses: Data Structure/Algorithms

B.M. in Music Performance, The University of Texas at Austin, USA *2015 - 2019*
- Completed all coursework in Computer Science, one course short of a minor.

Work Experiences

Summer Science Intern at Pandora/SiriusXM Summer 2022
- Research on methods of creating audio embeddings for downstream classification tasks
- Implemented the target task using such embeddings using CNN architectures for comparison of limitation in different tasks
- Extensively worked on Google Cloud Platform for developing and maintaining a ML infrastructure

Research Assistant at Academia Sinica, the National Academy of Taiwan *2019 - 2021*
- Researched on deep learning algorithms with PyTorch and TensorFlow for automatic music classification/auto-tagging, music generation, beat detection
- Published at three conferences and presented at several seminar talks

Academic Experiences

Peer-Reviewed Publications

[1] **Ching, J.**, Widmer, G. “Learning to Predict Performance-induced Emotion Differences in Classical Piano Music”, *International Society for Music Information Retrieval (**ISMIR**)*, 2026

[2] **Ching, J.**, Widmer, G. “A Study on the Data Distribution Gap in Music Emotion Recognition”, *17th International Symposium on Computer Music Multidisciplinary Research (**CMMR**)*, 2025

[3] Chou, Y-H., Chen, I-C., Chang, C-J., **Ching, J.**, Yang, Y-H. “BERT-like Pre-training for Symbolic Piano Music Classification Tasks”, *Journal of Creative Music Systems*, 2024

[4] Hung, H-T., **Ching, J.**, Doh, S-H., Kim, N., Nam, J., Yang, Y-H. “EMOPIA: A Multi-Modal Pop Piano Dataset for Emotion Recognition and Emotion-based Music Generation.”, *International Society for Music Information Retrieval (**ISMIR**)*, 2021

[5] Chiu, C-Y., **Ching, J.**, Hsiao, W-Y., Chen, Y-H., Su, W-Y., Yang, Y-H. “Source Separation-based Data Augmentation Techniques for Improved Joint Beat and Downbeat Tracking in Classical Music”. *European Signal Processing Conference (**EUSIPCO**)*, 2021

[6] **Ching, J.**, Ramires, A., Yang, Y-H. “Instrument Role Classification: Auto-tagging for Loop Based Music.” *Joint Conference on AI Music Creativity (2020)*. ISBN: 978-91-519-5560-5

[7] Ramires, A., Font, F., Bognadov, D., Smith, J. B. L., Yang, Y-H., **Ching, J.**,…Serra, X. “The FreeSound Loop Dataset and Annotation Tool.” *International Society for Music Information Retrieval (**ISMIR**)*, 2020

Non-refereed Publications

[8] **Ching, J.**, Cancino-Chacón, C. E., Widmer, G., “R3MO: A Multi-modal Dataset for Emotion Analysis in Orchestral Music", *International Conference on Computational and Cognitive Musicology (ICCCM)*, 2026

[9] Nikolakakis, E., **Ching, J.**, Karystinaios, E., Sipin, G., Widmer, G., Marinescu, R., “Language Models for Music Medicine Generation”, *Late-Breaking Demos of the 25th International Society for Music Information Retrieval Conference (ISMIR)*. arXiv preprint arXiv:2411.09080, 2024

[10] **Ching, J.**, Yang, Y-H. "Learning to Generate Piano Music with Sustain Pedals," *Late-Breaking Demos of the 22nd International Society for Music Information Retrieval Conference (ISMIR)*. *arXiv preprint arXiv:2111.01216,* 2021

[11] Chou, Y-H., Chen, I-C., Chang, C-J., **Ching, J.**, Yang, Y-H. "MidiBERT-Piano: Large-scale Pre-training for Symbolic Music Understanding". *arXiv preprint arXiv:2107.05223*, 2021

Talks

- 2026-05-09: Towards Understanding Emotion in Music, Applied Machine Learning Conference (ALMC), Poznań, Poland

Reviews

- International Society for Music Information Retrieval (2021, 2024 - present)
- Sound and Music Computing (2025 - present)
- IEEE Transactions on Affective Computing (TAFFC) (2025 - present)
- Transactions of the International Society for Music Information Retrieval (TISMIR) (2025 - present)
- Journal of New Music Research (JNMR) (2026)
- NeurIPS Workshop on AI for Music (2025)
- Latin American Music Information Retrieval (LAMIR) (2025 - present)

Research Projects

Analysis and Detection of Sustain Pedal Information *2022 - 2023*

Master project advised by Dr. Alexander Lerch, GaTech

- Analysis on the acoustic characteristics of multiple depths of the piano sustain pedal
- Apply time and frequency attention mechanism for the detection of pedal events in real-world recordings using a simple CNN architecture towards explainability
- Compared our implementations against various baselines
- In preparation for a journal submission

Pitch Tracking for Polyphonic Music *2021 Fall*

Research project advised by Dr. Claire Arthur, GaTech

- Leverage of my music theory knowledge for error analysis on existing works of f0 estimation
- Building new machine learning system based on the analysis results

Optimization of Guitar Tablature Transcription Using Beat Information *2021 Fall*

Audio Content Analysis course project supervised by Prof. Alexander Lerch, GaTech

- Proposed a system that incorporates beat information during data preprocessing to improve the transcription result
- Research on guitar pitch tracking/tablature transcription, and beat detection algorithm

Technical Skills

Machine Learning	Language: Python Tools: PyTorch, TensorFlow, Numpy, Scikit-learn, Matplotlib, Librosa,
Research	Music Information Retrieval, Machine Learning, Data Analysis, Statistics, Audio Signal Processing
Others	SQL, JavaScript, CSS, Git, HTML, Linux, Latex
Musical Skills	Violin, Piano, Percussion, Conducting, Music Theory, Music History

Awards

Dean's Fellowship, School of Music, Georgia Institute of Technology *2021 - 2022*

The Butler School of Music Scholarship, The University of Texas at Austin *2015 - 2019*

Other Experiences

Georgia Tech Symphony Orchestra (Concertmaster)	*2021 - 2022*
Philharmonia Moments Musicaux (Principal Second Violin)	*2020 - 2021*
Butler Opera Center Orchestra	*2019 April - May*
SXSW Audio Engineer	*2018 March*
University of Texas Symphony Orchestra	*2015 - 2019*
National Taiwan University Symphony Orchestra Concert Tour	*2017 Jul. - Aug.*